Data Structure and Algorithm Design

ME/MSc (Computer) – Pokhara University

Prepared by:

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Chapter 2:
Advanced Data Structures and Algorithms (11 hrs)

Outline

1. Advanced Trees
   - B and B+ Trees
2. Advanced Graph
   - Planar Graphs and its applications in Geographic and Circuit Layout Design
3. Network Flow Algorithms
   - Ford-Fulkerson Algorithm
   - Edmonds-Karp Algorithm
4. Advanced Shortest Path Algorithms
   - Bellman-Ford Algorithm
   - Johnson's algorithm
5. Graphical Data Structures
   - Quadtrees, KD Trees, R-Trees
6. Computational Geometry
   - Convex Hull and Voronoi Diagrams
7. Case Studies in Red-Black Tree and Its Applications, Applications of Directed Acyclic Graph, Applications of Minimum Spanning Tree in Network Design

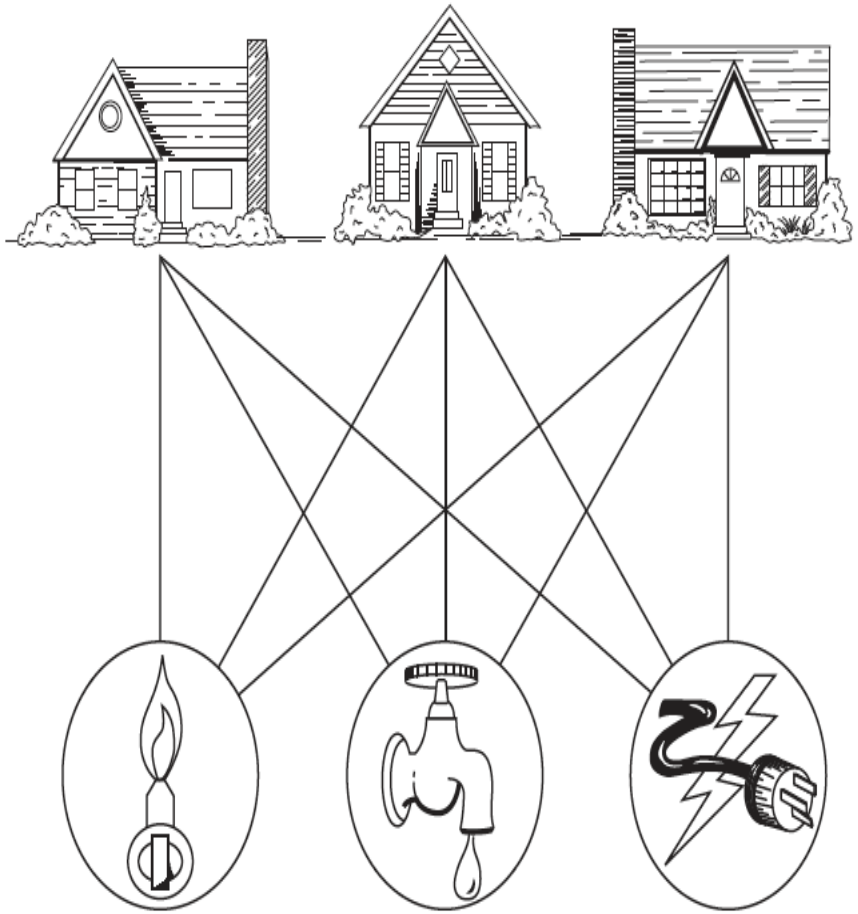
Advanced Graph

Advanced Graph: Planar Graphs and its applications in Geographic and Circuit Layout Design

Planar Graphs

- Consider the problem of joining three houses to each of three separate utilities as in Figure. Is it possible to join these houses and utilities so that none of the connections cross?

- There are always many ways to represent a graph. When is it possible to find at least one way to represent this graph in a plane without any edges crossing?



Planar Graphs

1. A graph is called **planar** if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a planar representation of the graph.
2. A graph may be planar even if it is usually drawn with crossings, because it may be possible to draw it in a different way without crossings.
3. A planar representation of a graph splits the plane into regions, including an unbounded region.
4. A complete graph of five vertices is non planar.

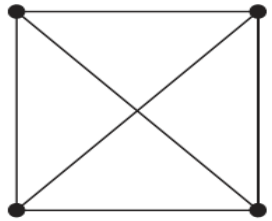


Fig: The Graph K4.

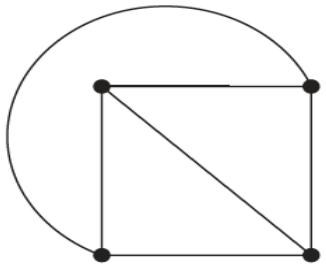


Fig: K4 Drawn with No Crossings.

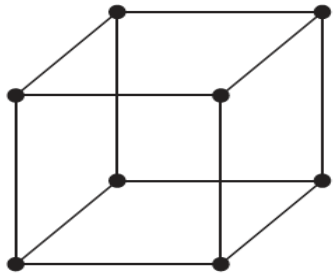


Fig: The Graph Q3.

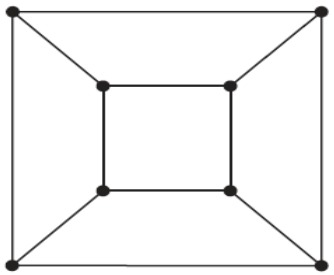
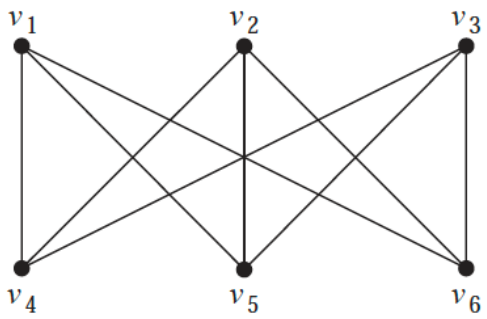
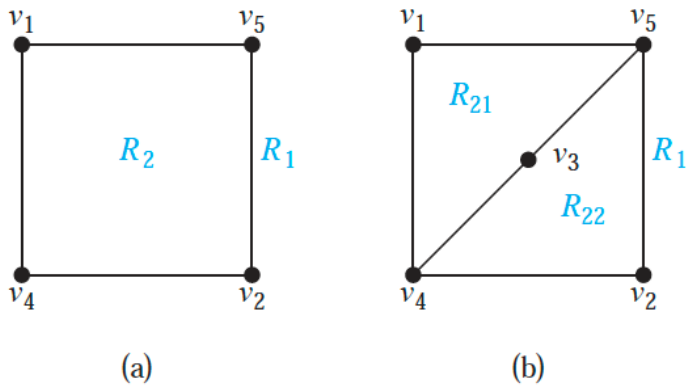


Fig: A Planar Representation of Q3.

Is $K_{3,3}$ graph(in figure shown) is planar?





- Any attempt to draw $K_{3,3}$ in the plane with no edges crossing is doomed.
- In any planar representation of $K_{3,3}$, the vertices v_1 and v_2 must be connected to both v_4 and v_5. These four edges form a closed curve that splits the plane into two regions, R_1 and R_2, as shown in Figure (a). The vertex v_3 is in either R_1 or R_2. When v_3 is in R_2, the inside of the closed curve, the edges between v_3 and v_4 and between v_3 and v_5 separate R_2 into two subregions, R_{21} and R_{22}, as shown in Figure (b).

Planar Graphs

1. **EULER'S FORMULA :** Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G. Then $r = e - v + 2$.

Poof:

We will use Mathematical induction

(a) Base Case:

For n=1 i.e. For number of edge = 1.

$r_1 = e_1 - v_1 + 2$

$r_1 = 1-2+2$

$r_1 = 1$ (TRUE)

(b) Induction Hypothesis:

Assume the equation is true for n=k i.e. for number of edges = k

$r_k = e_k - v_k + 2$ is true for G_k

EULER'S FORMULA

(c) Induction Step:

Let $\{a_{k+1}, b_{k+1}\}$ be the edge that is added to G_k to obtain G_{k+1}.

There are two case to be considered:

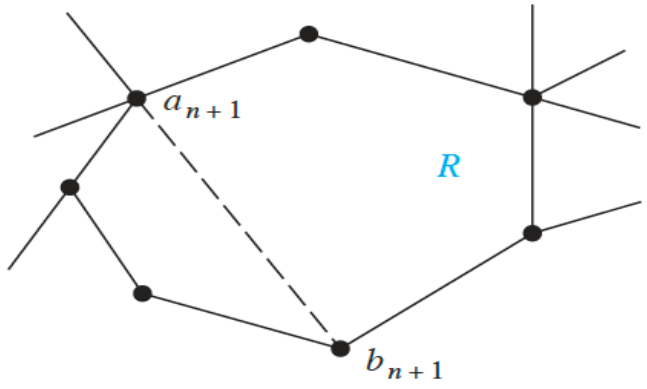
i. Both a_{k+1} and b_{k+1} are already in G_k (Fig a)

$$r_{k+1} = r_k + 1$$
$$e_{k+1} = e_k + 1$$
$$v_{k+1} = v_k \text{ \{Because both vertices are in } G_k\}$$
$$r_{k+1} = e_{k+1} - v_{k+1} + 2$$
$$r_k + 1 = e_k + 1 - v_k + 2$$
$$\mathbf{r_k = e_k - v_k + 2 \text{ which is true}}$$

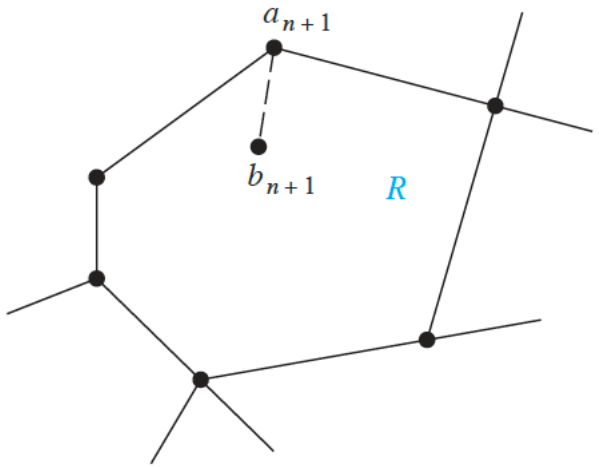
ii. One vertices is added to G_k (Fig b)

$$r_{k+1} = r_k$$
$$e_{k+1} = e_k + 1$$
$$v_{k+1} = v_k + 1 \text{ \{Because one vertices is added in } G_k\}$$
$$r_{k+1} = e_{k+1} - v_{k+1} + 2$$
$$r_k = e_k + 1 - (v_k + 1) + 2$$
$$\mathbf{r_k = e_k - v_k + 2 \text{ which is true}}$$

Hence by induction method Euler's Formula is proved



(a)



(b)

EULER'S FORMULA

Example

Q. 4 Suppose that a connected planar simple graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?

Solution:

This graph has 20 vertices, each of degree 3, so $v = 20$

Now using Handshaking theorem ,

$$2e = (20*3)$$

$$e = 30$$

Consequently, from Euler's formula, the number of regions is

$$r = e - v + 2$$

$$r = 30 - 20 + 2$$

$$r = 12.$$

Planar Graphs

Corollary of Euler's Theorem:

1. If G is a connected planar simple graph with e edges and v vertices, where $v \geq 3$, then $e \leq 3v - 6$

Proof:

Let G(v, e) be the connected planar simple graph. A connected planar simple graph drawn in the plane divides the plane into regions, say **r** of them.

The degree of each region is at least three. (Because the graphs discussed here are simple graphs, no multiple edges that could produce regions of degree two, or loops that could produce regions of degree one, are permitted.)

Degree of region is defined to be the number of edges on the boundary of this region.

Sum of the degrees of the regions is exactly twice the number of edges in the graph, because each edge occurs on the boundary of a region exactly twice . Because each region has degree greater than or equal to three, it follows that

$$2e = \sum_{all\ regions\ R} \deg(R) \geq 3r$$

$$(2/3)e \geq r$$

Using $r = e - v + 2$ (Euler's formula),

we obtain $v - e + r = 2$

$$v - e + (2/3)e \geq 2$$

$$\mathbf{e \leq 3v - 6.\ Hence\ proved}$$

Planar Graphs

- Show that K_5 is nonplanar using Corollary 1.

Solution:

The graph K_5 has five vertices and 10 edges.

However, the inequality $e \leq 3v - 6$ is not satisfied for this graph because

$$10 \leq 3*5 - 6$$

$10 \leq 9$ which is false. Therefore, K5 is not planar.

2. If G is a connected planar simple graph, then G has a vertex of degree not exceeding five

Proof:

If G has at least three vertices, by Corollary 1 we know that

$$e \leq 3v - 6,$$

$2e \leq 6v - 12$. [The sum of the degrees of all vertices in G is 2e]

$$2e + 12 \leq 6v$$

If the degree of every vertex were at least six, then because

$2e = \sum_{v \in V} \deg(v)$ (by the handshaking theorem), we would have

$$2e + 12 \geq 6v + 12.$$

But this contradicts the inequality $2e + 12 \leq 6v$. It follows that there must be a vertex with degree no greater than five.

Planar Graphs

3. If a connected planar simple graph has e edges and v vertices with $v \geq 3$ and no circuits of length three, then $e \leq 2v - 4$.

Proof: The sum of the degrees of the regions is equal to twice the number of edges. But each region must have degree ≥ 4 because we have no circuits of length 3. So we have $2e \geq 4f$. Then $(1/2)e \geq f$.

Euler's formula says that $v - e + f = 2$. or $e - v + 2 = f$.

Combining this with $(1/2)e \geq f$, we get

$e - v + 2 \leq (1/2)e$

So $(e/2) - v + 2 \leq 0$.

So $(e/2) \leq v - 2$.

Therefore $\mathbf{e \leq 2v - 4}$.

Planar Graphs

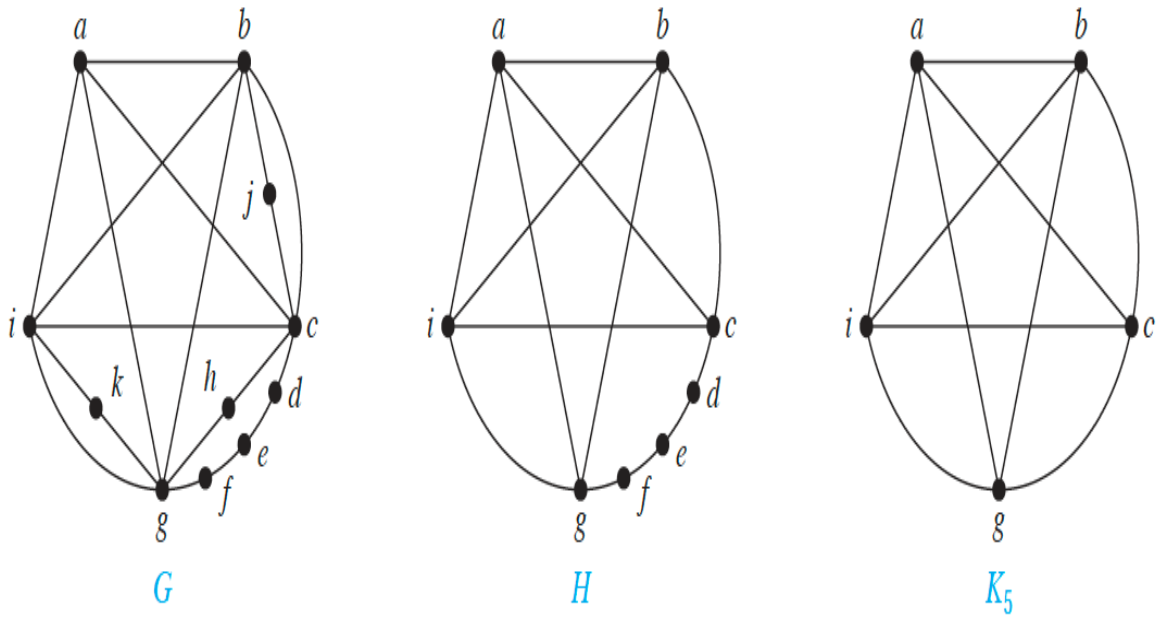
Kuratowski's Theorem: A graph is nonplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5.

[Two graphs are said to be homeomorphic if both can be obtained from the same graph by subdivisions of edges.]

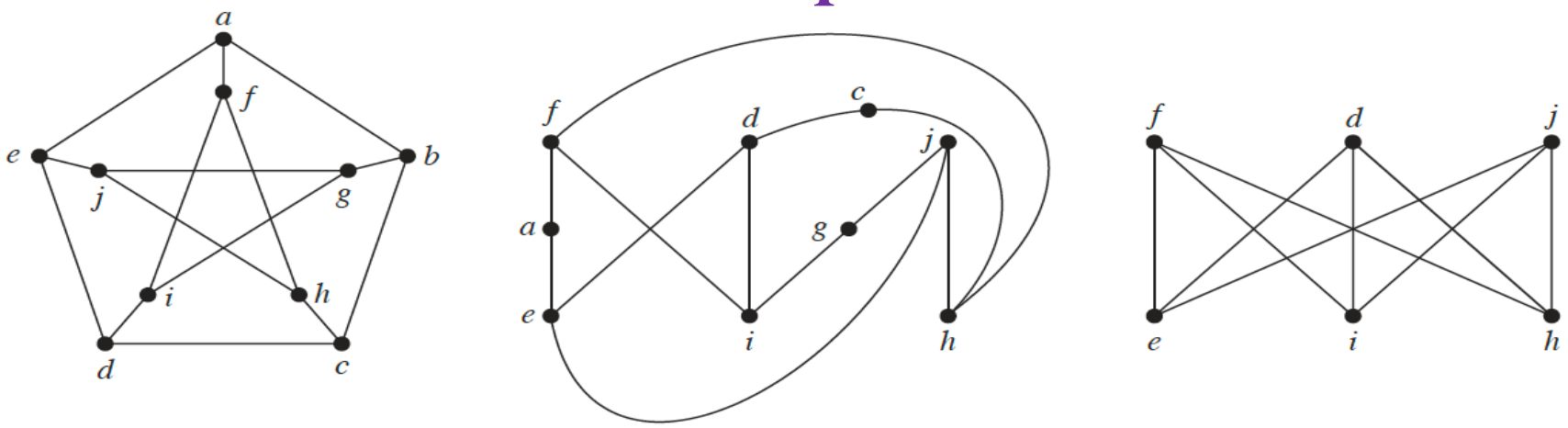
Proof: *G has a subgraph H homeomorphic to K5.*

H is obtained by deleting h, j , and k and all edges incident with these vertices. H is homeomorphic to K5 because it can be obtained from K5 (with vertices a, b, c, g, and i) by a sequence of elementary subdivisions, adding the vertices d, e, and f .

Hence, G is nonplanar.



Planar Graphs



(a) (b) H (c) $K_{3,3}$

- The subgraph H of the Petersen graph obtained by deleting b and the three edges that have b as an endpoint, shown in Figure (b), is homeomorphic to K3,3, with vertex sets {f, d, j} and {e, i, h}, because it can be obtained by a sequence of elementary subdivisions, deleting {d, h} and adding {c, h} and {c, d}, deleting {e, f } and adding {a, e} and {a, f }, and deleting {i, j } and adding {g, i} and {g, j}.
- Hence, the Petersen graph is not planar.

Planar Graphs

Applications of Planar Graph

- Planarity of graphs plays an important role in the design of electronic circuits. We can model a circuit with a graph by representing components of the circuit by vertices and connections between them by edges. We can print a circuit on a single board with no connections crossing if the graph representing the circuit is planar. When this graph is not planar, we must turn to more expensive options.
- For example, we can partition the vertices in the graph representing the circuit into planar subgraphs. We then construct the circuit using multiple layers.
- We can construct the circuit using insulated wires whenever connections cross. In this case, drawing the graph with the fewest possible crossings is important.
- The planarity of graphs is also useful in the design of road networks.
- Suppose we want to connect a group of cities by roads. We can model a road network connecting these cities using a simple graph with vertices representing the cities and edges representing the highways connecting them.
- We can built this road network without using underpasses or overpasses if the resulting graph is planar.

Assignment

Do some research and write a synopsis paper including case studies where planar graph can be use for Circuit layout design and GIS. Submit it in Google classroom.